

SMOOTHED SLICES

Complete AI Operator Guide for

SmoothedSlicesAppProgrammatic.m

Exact button sequences • Exact user-entered settings • Hidden automatic protocol settings
Scientific rationale • Interpretation rules • Export protocol

Version scope. This guide is written against the current project archive proj_zip2.zip (17 August 2026). The current MATLAB source is the authority for button names and available controls; the earlier project reports supply the research rationale. If a later build changes a control, the AI must reconcile the guide with the code before operating it.

Purpose: hand this document to an AI operator and require it to follow the sequence without inventing settings or conclusions.

1. Mission of the tool and the research question

The app is a comparative nonparametric-inference research instrument, not merely a visualization. Its purpose is to compare Slices, NF-iSAM, and the proposed Smoothed Slices method on shared cases, while isolating the question that motivated Smoothed Slices: can the nested sampling work that appears on the Lemma-1 / Eq.-(16) structural route be replaced by reusable conditional-expectation surfaces without sacrificing the posterior?

The conceptual step is the tower/smoothing property of conditional expectation: $E[Y] = E[E[Y | X]]$.

In the current project, Slices uses the nested inner estimator and Smoothed Slices uses the finite RCS (recursive conditional smoothing) surface estimator on the same elimination machinery. The research question is therefore not "which curve looks nicer?" It is whether Smoothed Slices can achieve matched posterior quality with less measured work, and whether the proposed reason for that saving—compact, sparse, reusable surfaces—actually holds.

Primary quantitative question. At matched posterior fidelity, does Smoothed Slices reduce measured factor evaluations relative to nested Slices, and if so is the change a genuine scaling improvement or only a constant-factor shift?

Quantity	Conceptual model	What the app must measure
Nested work	$C_{\text{nested}} \sim N^H R_s$	Measured factor evaluations + posterior error across budget
Smoothed work	$C_{\text{RCS}} \sim \sum_r B_r K_r R_s$	Measured factor evaluations, $ X_r $, $ N_r $, $ S $, surface diagnostics
Posterior fidelity	Must not be traded away for speed	Relative L1 error, MMD, RMSE, mode weights / collapse where relevant
Surface premise	R_r should be compact / sparse / approximable	Singular-value decay, rank, sparsity, active-set error, retained mass
Practical runtime	Implementation-dependent secondary evidence	Wall time and phase timings; never substitute this for the work counter

1.1 What is paper-derived and what is our proposed extension

- **Slices:** baseline direct slice-based nonparametric inference. In the current code its driver requests the nested inner estimator.
- **NF-iSAM:** a separate normalizing-flow / Bayes-tree comparison method. It is essential for posterior-recovery context, but it does not share the same factor-evaluation cost unit used by the Slices-vs-Smoothed research axis.
- **Smoothed Slices:** our proposed extension. It retains the Slices factor graph and elimination structure but replaces the nested inner estimator with finite conditional-smoothing surfaces on the structural route that actually needs it.
- **Current implemented RCS mode:** finite-support Mode A. Basis/regression/proposal/hybrid ideas belong to the research roadmap; they are not selectable in the present GUI.

2. Rules the AI operator must obey

Rule	Operational meaning
Current code is the UI authority.	Never invent a button, slider, mode, or hyperparameter because an older specification mentions it.
Use appFast only for smoke tests.	No number from appFast may be reported as a scientific result.
Run the batch reference first.	Leave incremental replay OFF for the primary comparisons. Incremental replay is a separate experiment.
Do not use wall time as the primary cost proof.	The Research Question tab deliberately uses measured factor evaluations; wall time is hardware/MATLAB dependent.
Do not put NF-iSAM on the factor-evaluation frontier.	Its sampling semantics make the count incomparable on that axis. Use NF-iSAM for posterior-recovery/runtime context instead.
Do not claim Smoothed Slices is exercised on every case.	In this current build, the direct nested-vs-RCS distinction is exercised by the Two-pose range / Eq.-16 route. Four Doors, Grid World, and Plaza use the general engine and therefore act as posterior/robustness cases, not direct proof of the smoothing mechanism.
Always compare quality and work together.	A lower-cost answer that has lost modes or increased posterior error is not a computational win.

Rule	Operational meaning
Record the seed and every visible budget.	The dataset/noise stream and estimator budget are intentionally separated; reproducibility requires both.
Treat warnings as data.	Do not hide retained-mass warnings, support issues, engine failures, or method deviations.
Export before replacing an important single-run result.	A new Run All/Run Selected replaces app.Results. The sweep/research tabs keep separate in-memory result objects, but the generic numerical export bundle is centered on app.Results.

Critical current-build boundary. The app itself states that the Two-pose benchmark is the only case whose elimination takes the Lemma-1 route and the only current case that builds the conditional smoothing surface used by the RCS research studies. Do not use equality/difference between Slices and Smoothed Slices on Four Doors, Grid World, or Plaza as evidence for or against the surface recursion.

3. Launch and preflight

P0 Launch the current app

Evidence role: *Reproducibility and version control; no inference yet.*

P0.1 Open the project

SET	Open the extracted proj_zip2 project in MATLAB. The launch function is located under src/ and resolves its own path.
PRESS	From MATLAB run: <code>app = runSmoothedSlicesApp();</code>
INSPECT	The window title must read “Smoothed Slices - comparative posterior recovery”. Confirm the tab strip includes Reproducibility, Case Study, Compare Methods, Posterior Recovery, Process Explorer, Plaza, Sweep, Research Question, Surface Complexity, Figure Generator, Diagnostics.
WHY	This verifies that the AI is operating the current programmatic frontend rather than an older App Designer build.
ACCEPT / RECORD	Record MATLAB release and platform shown on the Reproducibility tab.

```
app = runSmoothedSlicesApp();
```

P0.2 Reproducibility preflight

SET	Reproducibility tab: leave Random Seed = 0 for the smoke test. Read the results and presentation output folders.
PRESS	No run button yet.
INSPECT	Confirm RNG note = “threefry, one stream per case”; confirm output folders exist or were created this session.
WHY	The project separates the data/noise stream from method budgets. Changing a sample count must not silently redraw the data.
ACCEPT / RECORD	If folders are missing or project health is abnormal, stop scientific operation and fix the repository before generating evidence.

4. Exact control inventory

This section is the operator’s map. The wording below follows the current source exactly.

Tab	Control	Meaning
-----	---------	---------

Tab	Control	Meaning
Reproducibility	Random Seed	Estimator / stochastic-method seed for ordinary single runs.
Case Study	case: Two-pose range / Four Doors / Grid world / Plaza1 / Plaza2	Loads the problem; changing it clears current single-run results.
Case Study	variant: gaussian / multimodal	Two-pose: Gaussian vs ambiguous/multimodal factor. Grid/Plaza: known vs ambiguous association. Disabled for Four Doors.
Case Study	layout: office / corridor / warehouse	Grid World only. Changing layout also resets pose count to that layout's baseline.
Case Study	poses	Grid World only; enabled range 3–16.
Case Study	budget preset: paperLike / appFast	Changes both visible budgets and hidden NF-iSAM training budget.
Case Study	X_0 outer samples	Outer Slices-family sample budget.
Case Study	L_n nested inner samples	Nested Slices inner budget; quantity the RCS proposal seeks to avoid.
Case Study	S separator support	Finite separator support cardinality.
Case Study	X_1 RCS path support	Finite path support for the conditional surface.
Case Study	N_0 active successors	Sparse active-set size. Setting it equal to X_1 selects the dense update in current UI defaults.
Case Study	backward separator samples	Posterior/backward sampling budget.
Case Study	marginal grid	Grid used for marginal display/normalization.
Case Study	replay increment by increment	OFF by default; enables the two replay controls.
Case Study	reuse surfaces across increments	Incremental reuse toggle; meaningful only when replay is ON.
Case Study	MMD early stopping (Alg. S5)	Backward traversal stopping heuristic; meaningful only when replay is ON.
Compare Methods	include checkboxes + Run Selected	Runs only checked methods for an ordinary single case.
Posterior Recovery	Run All	Runs Slices, NF-iSAM, Smoothed Slices regardless of Compare Methods checkboxes.
Posterior Recovery	Export All Figures + Diagnostics	Exports the generic registered-figure/numerical bundle after a single run.
Process Explorer	increment k / stage / method / Play / Export frame	Replays the internal process after a run; structural stepping is also available before a run.
Sweep	plan dropdown + Run Sweep	Runs an internally defined multi-cell experiment. Compare Methods checkboxes do not select sweep methods.
Research Question	Measure cost vs quality	Runs only Slices + Smoothed Slices on the controlled Two-pose frontier.
Surface Complexity	Measure surface complexity	Runs the 17-configuration E2 screening design with its own internal budgets.
Surface Complexity	Profile the active set	Runs the active-K profile with its own internal budgets.
Plaza	landmark ordinal L + Run evidence ladder (P2-A)	Runs three independent Plaza evidence windows using the methods currently checked on Compare Methods.
Figure Generator	PNG and PDF / PNG only / PDF only; Refresh checklist; Export bundle	Audits drawn/empty registered axes and exports the current bundle.
Diagnostics	read-only tables/warnings	Runtime, quality, replay status, warnings. No run controls.

5. Budget presets and exact values

Visible field	paperLike	appFast	Operator interpretation
---------------	-----------	---------	-------------------------

Visible field	paperLike	appFast	Operator interpretation
X_0 outer samples	200	60	Primary outer sampling budget
L_n nested inner samples	50	20	Nested inner work; Slices-specific mechanism budget
S separator support	201	61	Separator finite support
X_1 RCS path support	200	60	Surface path support
N_0 active successors	200	60	Displayed finite equivalent of default dense branch
backward separator samples	400	60	Backward/posterior sampling
marginal grid	401	101	Marginal display/evaluation grid
Hidden NF-iSAM training samples	2000	500	Changed by preset; not visible as an edit field
Hidden NF-iSAM iteration cap	200	60	Changed by preset; not visible as an edit field
Hidden MMD evaluation samples	1000	200	Scoring budget; not the 100-sample early-stop budget

Why the preset matters. Do not make a run “fast” by lowering only |X_0|. NF-iSAM training and the backward pass can remain dominant. The appFast preset moves the budget group coherently and stamps the preset name into exported configuration. Its purpose is UI testing, not reported science.

6. Master experimental sequence

Order	Protocol	Exact route	Why it exists
0	Smoke test	Two-pose Gaussian, appFast, all three	Confirm wiring/UI/exports; never report result
1	Canonical Eq.-16 Gaussian	Two-pose Gaussian, paperLike, seed 3	Baseline exact-quadrature mechanism case
2	Canonical Eq.-16 multimodal	Two-pose multimodal, paperLike, seed 3	Mode-preservation stress on same structural route
3	Cost-vs-quality frontier	Research Question button	Primary answer to computation-vs-quality question
4	Surface screening	Measure surface complexity	Test compactness premise
5	Active-set profile	Profile the active set	Test how small K can be and whether predicted saving is realized
6	Four Doors baseline + sweep	paperLike / Four Doors plan	Posterior-shape and mode-collapse external case
7	Grid World baseline + sweep	office baseline + problem-size plan	General-engine scaling/geometry external case
8	Plaza1 / Plaza2 baselines	paperLike with N=150, seed 7	Real-data windowed posterior recovery
9	Plaza evidence ladder	Plaza2, L=2, P2-A	Landmark marginal evolution across independent evidence windows
10	Optional incremental replay	Same ordinary case; batch first, replay second	Reuse/early-stop exploratory behavior; not a substitute for batch reference
11	Process capture + exports	Process Explorer + Figure Generator	Preserve evidence and implementation trace

7. Protocol 0 — smoke test only

T0 UI and method smoke test

Evidence role: *Functional validation only; explicitly non-scientific.*

T0.1 Configure

SET	Reproducibility: Seed = 0. Case Study: case = Two-pose range; variant = gaussian; budget preset = appFast. Verify visible values: X_0 =60, L_n =20, S =61, X_1 =60, N_0 =60, backward=60, marginal grid=101. Leave replay increment by increment OFF.
PRESS	Posterior Recovery → Run All.
INSPECT	All three methods should complete or report an explicit method status. Generated-factor/reference, marginal, joint samples, surface panel, and metrics/cardinality panels should draw.
WHY	The only purpose is to prove that the frontend, method dispatch, reference generator, plotting, and progress/Stop plumbing work before a long experiment.
ACCEPT / RECORD	Mark this run “SMOKE / appFast / NOT FOR REPORTING”.

Do not misread this run. Never compare appFast accuracy or NF-iSAM flow quality with paperLike results.

8. Protocol 1 — canonical Two-pose Gaussian benchmark

E16-G Two-pose range, Gaussian

Evidence role: Direct exact-quadrature test of the structural route where Slices nested and RCS differ.

E16-G.1 Set the canonical configuration

SET	Reproducibility: Seed = 3. Case Study: case = Two-pose range; variant = gaussian; budget preset = paperLike. Set/verify X_0 =200, L_n =50, S =201, X_1 =200, N_0 =200, backward=400, marginal grid=401. Leave replay OFF. Compare Methods: leave Slices, Smoothed Slices, and NF-iSAM all checked.
PRESS	Posterior Recovery → Run All.
INSPECT	Read Generated factor versus exact quadrature, Marginal, Conditional smoothing surface, Joint posterior samples, and Metrics/cardinalities. Then open Diagnostics and read warnings.
WHY	This is the smallest case with an exact reference and elimination order (x1, l1, x2). The l1 variable has no unary factor, so the Lemma-1 structural route is actually exercised rather than inferred.
ACCEPT / RECORD	Record for each method: status, relative L1 error, MMD, runtime; for Slices/Smoothed also factor evaluations and cardinality diagnostics. Record any warnings verbatim.

E16-G.2 Inspect the mechanism, not only the final posterior

SET	Process Explorer: increment k = 1; stage = 2; method = Slices. Then repeat method = Smoothed Slices. Use Compare for the side-by-side context if desired.
PRESS	No new inference button. Move the stage slider; use Export frame once the l1 elimination stage is visible.
INSPECT	Confirm stage 2 is the l1/no-unary elimination in the current process card. Slices should expose the nested inner construction; Smoothed Slices should expose the finite conditional-smoothing-surface construction.
WHY	This checks that the two method labels correspond to the intended estimator route. A cost comparison is meaningless if both traces accidentally took the same route.
ACCEPT / RECORD	Save an Export frame for Slices stage 2 and another for Smoothed Slices stage 2.

Do not misread this run. If stage 2 does not identify l1 in a later code build, locate the stage whose eliminated variable is l1 before interpreting it; do not blindly interpret the numeric index.

9. Protocol 2 — canonical Two-pose multimodal benchmark

E16-M Two-pose range, multimodal

Evidence role: Same direct mechanism test, now with ambiguity that can punish over-smoothing or mode collapse.

E16-M.1 Change only the posterior structure

SET	Keep Seed = 3 and all paperLike budgets exactly as in E16-G. Change only Case Study → variant = multimodal. The current case builder automatically uses mixture offset d=2.2 and weak odometry sigma=3.0 for this variant. Replay remains OFF.
PRESS	Posterior Recovery → Run All.
INSPECT	Compare the exact-reference generated factor, marginal modes, sample cloud, relative L1 error and MMD. Inspect the Smoothed surface for loss of one branch/mode.
WHY	Holding budgets and seed fixed makes the experiment about posterior structure, not spending. The multimodal variant is the support-preservation stress test on the same structural route.
ACCEPT / RECORD	Record mode preservation qualitatively and the same quantitative metrics as E16-G. Export before changing cases.

10. Protocol 3 — the primary Research Question measurement

RQ Measure cost vs quality

Evidence role: Primary quantitative answer to whether Smoothed Slices reduces measured work at matched posterior quality.

Important. This button ignores the Compare Methods method checkboxes and deliberately runs only Slices and Smoothed Slices. That is correct. NF-iSAM has no honest point on the measured factor-evaluation axis used here.

RQ.1 Normalize the unswept settings

SET	Before pressing the research button, return Case Study to budget preset = paperLike and verify the standard visible values L_n =50, S =201, X_1 =200, N_0 =200, marginal grid=401. The research plan itself overrides X_0 , backward samples, and seed in each cell.
PRESS	Research Question → Measure cost vs quality.
INSPECT	After completion, use the scenario dropdown to inspect every scenario. Read both plots and the one-verdict-per-scenario table.
WHY	The plan varies posterior sampling budget while holding the non-swept machinery stable. This isolates how work and error scale together rather than comparing one arbitrary point.
ACCEPT / RECORD	For every scenario, record the Slices and Smoothed work-vs-budget trend, posterior-error frontier, fitted growth-rate conclusion/verdict, and whether the mechanism was exercised.

Automatic setting inside the RQ button	Value
Two-pose separation set	d = 2.2 only
Budgets	N = 50, 100, 200, 400
Budget coupling	numSamples = N and numBackwardSamples = N
Seed	3
Scenarios	4: unimodal tight; unimodal weak; d=2.2 weak; d=2.2 tight

Automatic setting inside the RQ button	Value
Methods	Slices + Smoothed Slices only
Primary x-axis	Measured factor evaluations
Primary y-axis	Posterior error
Secondary plot	Work vs budget / growth rate

Decision rule. A lower curve is not automatically a win. First require comparable posterior quality, then compare measured factor evaluations. If both methods have the same growth exponent but one uses fewer evaluations, report a constant-factor reduction—not a change of complexity class. If the error is worse at the same work, do not call it a saving.

11. Protocol 4 — surface-complexity screening

E2-S Measure surface complexity

Evidence role: Tests the premise behind the proposed saving: are the R surfaces actually compact/structured?

E2-S.1 Run the one-axis-at-a-time screening design

SET	No Case Study budget edit is required for this button; it intentionally uses its own internal study budget. Do not change visible fields expecting them to affect this study.
PRESS	Surface Complexity → Measure surface complexity.
INSPECT	After completion, cycle the axis dropdown through every available numeric axis, especially control, samplingNoise, fusionNoise, and overlap. Read singular-value decay, rank/sparsity panel, table, verdict, inherited-rank note, and control status.
WHY	This asks whether the RCS representation has lower effective dimension/structure than the raw nested sample tree. It also contains a negative control: odometry sigma is outside the surface recursion and therefore should not move the spectrum.
ACCEPT / RECORD	Record compactness verdict, energy rank / relative rank, terminal-vs-R0 rank relationship, sparsity, and whether the control stayed exactly flat.

Automatic E2 screening parameter	Values / behavior
numSamples / X ₀	120
X ₁ surface support	120
S separator support	101
backward samples	100
MMD evaluation samples	100
active set K	24
Variants	gaussian and multimodal
Control: odometry sigma	0.3, 1.2, 3.0
Sampling-factor g ₀ sigma	0.2, 0.5, 1.0, 2.0
Fusion-factor a sigma	0.2, 0.5, 1.0, 2.0
Multimodal overlap / mixture offset	0.8, 1.5, 2.2, 3.5
Design	17 runs, one axis at a time; not a factorial design

Automatic E2 screening parameter	Values / behavior
Nonlinearity	Not swept in current build; nonlinear general-engine cases do not build this surface

Interpret rank carefully. Low numerical rank, sparsity, and small active K are different claims. Rank 1 is not automatically a success: in this app it can mean the surface has become separable and the conditioning carries no useful information. Also compare R0 rank with terminal R1 rank; if R0 merely inherits R1's low rank, do not attribute that compactness to smoothing.

12. Protocol 5 — active-successor profile

E2-K Profile the active set

Evidence role: Determines how small $|N0|$ can be before the surface or final posterior changes, and checks whether theoretical sparse cost is actually realized.

E2-K.1 Run the K profile

SET	No visible Case Study budget edit is required; this button uses its own fixed study budget and the multimodal Two-pose case.
PRESS	Surface Complexity → Profile the active set.
INSPECT	Read the active-set error plot and the headline. Distinguish the K required to match the dense surface from the K required to match the dense posterior answer.
WHY	The surface can change noticeably while the final posterior remains within the sampling floor. Reporting only one K would hide that distinction.
ACCEPT / RECORD	Record both recommended K values, their fraction of $ X_1 $, retained mass, union fraction, predicted cost, measured factor evaluations, posterior error, and the cost-realized verdict.

Automatic active-set setting	Value
Case	Two-pose multimodal
$ X_0 $	120
$ X_1 $	120
$ S $	101
backward samples	100
K values	2, 4, 8, 16, 32, 64, 128, plus dense K = infinity
Surface matching tolerance	1e-3 relative Frobenius error
Reference	Dense K = infinity surface + quadrature posterior reference

Most important cost-reading rule. The current active-set implementation can show a strong reduction in the predicted sparse cost while measured factor evaluations remain unchanged, because transition weights are built on the full support before sparsification. If that happens, report: “the representation is sparse in the cost model, but the present implementation has not realized the evaluation saving.” Never quote predictedCost as measured speedup.

13. Protocol 6 — Four Doors posterior-shape evidence

FD-1 Four Doors single run

Evidence role: *External multimodal posterior test; not a direct RCS-mechanism test in the current general engine.*

FD-1.1 Canonical single run

SET	Reproducibility Seed = 3. Case Study: case = Four Doors. Variant is disabled. Set budget preset = paperLike and verify default visible fields. Replay OFF. Compare Methods: all three checked.
PRESS	Posterior Recovery → Run All.
INSPECT	The Generated factor panel is repurposed to mode weights; the Marginal panel shows chain marginals. Read mode preservation/collapse before RMSE.
WHY	Four Doors exists because an estimator can lose a live mode yet improve RMSE by landing closer to the truth. Posterior shape is therefore the evidence, not only the mean error.
ACCEPT / RECORD	Record worst/visible mode-weight errors, marginal shape, collapsed modes if reported, MMD/RMSE, runtime, and warnings.

Do not misread this run. Do not interpret any equality between Slices and Smoothed Slices here as a result about RCS; the general engine does not exercise the inner-estimator difference.

FD-S Four Doors sweep

Evidence role: *Trend test across budget, door spacing, odometry, symmetry/asymmetry, and corridor length.*

FD-S.1 Run the automatic plan

SET	Return Case Study budget preset to paperLike before the sweep so unswept settings are standardized. No need to select Four Doors on the Case Study tab; the Sweep plan builds its own cases.
PRESS	Sweep → plan = Four Doors: spacing and odometry → Run Sweep.
INSPECT	Read the six panels. Start with mode-weight L1 and collapsed modes. Read the irregular-door scenario before drawing a conclusion from symmetric corridors. Use RMSE only as a secondary, potentially flattering metric.
WHY	The irregular corridor breaks symmetry, preventing a collapsed estimator from accidentally landing near truth. The tight/weak odometry pair separates ambiguity from resolvability.
ACCEPT / RECORD	Record scenario, budget, method, worst mode-weight L1, mean marginal L1, collapsed modes, RMSE, runtime.

Automatic Four Doors sweep	Values
Budgets	50, 100, 200, 400
Door spacings	2.5, 4, 6
Weak odometry sigma	0.55
Tight odometry sigma	0.18
Additional scenarios	irregular [2,6,8,14]; six doors / eight poses
Seed	3 for data and methods
Cells	6 scenarios x 4 budgets = 24; all three methods

14. Protocol 7 — Grid World general-engine evidence

GW-1 Grid World baseline

Evidence role: General nonlinear range-only case for geometry/problem-size behavior; not direct proof of RCS.

GW-1.1 Known-association office baseline

SET	Reproducibility Seed = 11. Case Study: case = Grid world; variant = gaussian; layout = office. Changing layout automatically sets poses = 5 for office. Set budget preset = paperLike. Replay OFF. All three methods checked.
PRESS	Posterior Recovery → Run All.
INSPECT	Read map, pose/landmark recovery, engine health, sample cloud, cardinalities, and Diagnostics warnings.
WHY	This establishes the standard known-association general-engine case at the seed used by the structural sweep.
ACCEPT / RECORD	Record number of variables shown in the case note, maximum separator dimension, pose RMSE, landmark distribution diagnostics, support/ESS health, runtime and warnings.
Do not misread this run. Do not turn the app's office-specific "about 13" annotation into a universal engine limit.	

GW-1.2 Ambiguous-association partner

SET	Keep Seed = 11, layout = office, poses = 5, paperLike budgets. Change only variant = multimodal (this maps to ambiguous association in Grid World).
PRESS	Posterior Recovery → Run All.
INSPECT	Compare posterior shape/health against GW-1.1 and inspect warnings. Keep the same interpretation boundary: this is a general-engine robustness case.
WHY	This changes association ambiguity while holding map/mission/budget fixed.
ACCEPT / RECORD	Record the change in posterior-quality and engine-health metrics relative to the known-association partner.

GW-S Grid World problem-size sweep

Evidence role: Tests whether failure/quality trends are geometry-specific rather than a universal variable-count threshold.

GW-S.1 Run the automatic structural plan

SET	Return budget preset = paperLike. The sweep overrides N=200, backward=200, S =250 and per-cell seed; unswept settings come from the current base config.
PRESS	Sweep → plan = Grid world: problem size → Run Sweep.
INSPECT	Read pose RMSE vs variables with marker identity retained. Compare office, corridor, warehouse, extra seeds, sensing range, and ambiguous-association cells. Do not average away layout identity.
WHY	The sweep was specifically built to falsify the earlier assumption of one universal variable-count cliff. Geometry and mission length can shift where quality degrades.
ACCEPT / RECORD	Report turns/failures per layout and seed, not one engine-wide safe variable count.

Automatic Grid World sweep family	Cells
Size	Office, 4–7 poses (4 cells), base seed 11

Automatic Grid World sweep family	Cells
Seed	Office 5 and 6 poses at seeds 3, 7, 13 (6 cells)
Layout	Corridor and warehouse at 5 and 7 poses (4 cells)
Sensing	Office range 4.5/5.5/6.5 m + ambiguous association (4 cells)
Budget	N=200, backward=200, S =250
Total	18 cells, all three methods

15. Protocol 8 — Plaza real-data baselines

Plaza is a windowed real-data case in this implementation. Treat each window as an independent posterior, not as a full-sequence reproduction of the papers. The app uses calibrated range data, a 30 m range cutoff, eight keyframes by default subject to a variable cap, and landmarks-last elimination.

P1-A(UI) Plaza1 known-association baseline

Evidence role: *Real-data posterior recovery on the current windowed engine.*

P1-A(UI).1 Configure a paper-style ordinary run

SET	Reproducibility Seed = 7. Case Study: case = Plaza1; variant = gaussian. Select budget preset = paperLike, then manually set X_0 outer samples = 150. Keep L_n =50, S =201, X_1 =200, N_0 =200, backward=400, marginal grid=401. Replay OFF. All three methods checked.
PRESS	Posterior Recovery → Run All.
INSPECT	Read the Plaza context figure at the scale of the full run; inspect pose recovery, sample distribution, and landmark marginals/distribution. Read aligned pose RMSE and warnings.
WHY	The Plaza protocol documentation uses N=150; the repository's generic paperLike default is N=200. The manual override makes the visible ordinary run closer to the Plaza paper-style sampling budget while retaining hidden NF-iSAM train samples = 2000.
ACCEPT / RECORD	Record dataset/window keyframes, number of variables/readings/landmarks, range sigma after calibration, pose RMSE (aligned and raw if available), posterior shape, runtime, warnings.
Do not misread this run. Do not judge a bimodal range-only landmark posterior by its mean alone; the mean may sit between two valid reflected modes and explain the ranges poorly.	

P2-A(UI baseline) Plaza2 known-association baseline

Evidence role: *Same real-data windowed comparison on Plaza2 before running the evidence ladder.*

P2-A(UI).1 Change dataset only

SET	Keep Seed = 7 and exactly the same Plaza paper-style budgets used for Plaza1. Change Case Study → case = Plaza2; variant = gaussian.
PRESS	Posterior Recovery → Run All.
INSPECT	Read the context figure and distributional diagnostics before the landmark evidence ladder.
WHY	This holds method settings fixed while changing the real dataset.
ACCEPT / RECORD	Export/record the baseline before starting the ladder.

16. Protocol 9 — Plaza evidence ladder (P2-A button)

P2-A Landmark evidence ladder

Evidence role: Shows how one landmark marginal changes across three independent windows with increasing evidence.

P2-A.1 Prepare the method selection and settings

SET	Case Study must currently be Plaza2, variant gaussian. Reproducibility Seed = 7. Use paperLike with X_0 =150 and the remaining Plaza values from Protocol 8. Compare Methods: check Slices, Smoothed Slices, and NF-iSAM for the canonical comparative ladder. Plaza tab: landmark ordinal L = 2.
PRESS	Plaza → Run evidence ladder (P2-A).
INSPECT	Read the six marginal panels: x on the upper row, y on the lower row, evidence increasing left-to-right. Read the Ladder metrics table.
WHY	The original paper-style step numbers do not map cleanly onto this implementation's 6 m keyframes. The code therefore chooses three windows from the actual data: few, middle, and most observing poses for the selected landmark. These are independent posteriors.
ACCEPT / RECORD	Record each rung's window, observing-pose count, method metrics, and marginal shape. Explicitly state that rungs are independent windows, not one recursive filter.

Method-selection behavior. Unlike Sweep/Research/Surface buttons, the Plaza ladder DOES use the checkboxes on Compare Methods. If the scientific question is only Slices vs Smoothed Slices, you may run a secondary two-method ladder by unchecking NF-iSAM; label it as such. The canonical comparative ladder in this guide keeps all three.

17. Optional robustness run — ambiguous Plaza association

PL-A.1 Controlled ambiguity partner

SET	Load Plaza1 or Plaza2; keep Seed = 7 and the same budgets as its known-association baseline. Change only variant = multimodal. In the current Plaza builder this maps to ambiguous association with default ambiguity fraction 0.5.
PRESS	Posterior Recovery → Run All (or Run Selected if a specifically labelled method subset is intended).
INSPECT	Inspect distributional changes, engine warnings, and association-related degradation. Compare against the corresponding gaussian/known run.
WHY	This is a UI-exposed ambiguity stress test. It is not the full P1-B 0/20/40/60% ambiguity sweep from the script-level protocol.
ACCEPT / RECORD	Record that UI multimodal Plaza uses the builder's default 50% ambiguity; do not describe it as a fraction sweep.

18. Optional incremental replay / reuse study

Exploratory status. The batch path is the project's reference and should be reported first. Incremental replay, surface reuse, and MMD early stopping are valuable implementation diagnostics, but the current GUI does not constitute a complete E6 mechanistic ablation proving the Smoothed-Slices saving. Use this section as a controlled replay study, not as the primary research conclusion.

INC.1 Full replay features

SET	Choose an ordinary case with multiple increments (Grid World or Plaza). Keep its baseline seed/budgets unchanged. Case Study: tick replay increment by increment. Leave reuse surfaces across increments = ON and MMD early stopping (Alg. S5) = ON. For a Slices-family-focused replay, Compare Methods: check Slices + Smoothed Slices and uncheck NF-iSAM.
PRESS	Compare Methods → Run Selected.
INSPECT	Diagnostics → Incremental replay and early stopping. Process Explorer → move increment k through the mission.
WHY	This tests whether the implementation reuses generated objects and stops backward traversal when the configured MMD criterion is met.
ACCEPT / RECORD	Record cache/reuse indicators, early-stop behavior, per-increment runtime/quality, and warnings.

INC.2 Reuse ablation

SET	Keep replay ON and all other settings fixed. Turn reuse surfaces across increments = OFF; leave early stopping ON.
PRESS	Run Selected.
INSPECT	Compare against INC.1. Then, if needed, run a separate partner with cache ON / early stop OFF. Never change two toggles and the dataset/budget simultaneously.
WHY	One-control-at-a-time changes identify which replay mechanism changed the result.
ACCEPT / RECORD	Label these as replay-ablation results, not as proof that the current Two-pose RCS surface itself produced the saving.

19. Process Explorer operating instructions

Control	Exact use	Scientific role
increment k	Select which data increment is being displayed.	Separates incremental time from elimination stage.
stage	Select elimination/process stage within that increment.	Shows current variable/cliue/surface operation.
method	Compare / Slices / Smoothed Slices / NF-iSAM	Changes the process trace being explained.
Play	Animate the current process timeline.	Educational/review aid; not a new inference run.
Export frame	Save the currently displayed process frame.	Creates evidence for reports/presentations of the algorithmic distinction.

For the central Two-pose benchmark, the most important frame is increment k=1, stage=2, where l1 is eliminated. Use the stage card to verify the eliminated variable rather than trusting the number alone if the code is revised. Capture one Slices frame and one Smoothed Slices frame under the same case/budget.

20. Export and evidence-preservation protocol

EXP Export the current evidence

Evidence role: Makes the run auditable and prevents later runs from silently replacing the state you intended to report.

EXP.1 Export immediately after each canonical ordinary run

SET	Posterior Recovery must contain the run you intend to preserve. Figure Generator: Image format = PNG and PDF unless a narrower output is explicitly required.
PRESS	Posterior Recovery → Export All Figures + Diagnostics, or Figure Generator → Refresh checklist → inspect states → Export bundle.
INSPECT	Refresh checklist must distinguish drawn axes from “empty, not yet run”. Empty registered axes still export as blank images.
WHY	The figure checklist prevents a folder of blank axes from being mistaken for failed exports and makes missing experiments visible before the export.
ACCEPT / RECORD	Confirm the output folder contains numerical and image artifacts; keep the configuration associated with the run.

Bundle file	Meaning
run_state.mat	Raw ordinary app.Results plus configuration passed to the generic exporter
metrics.csv	Comparison table from the ordinary run
config.json	Settings, seed, MATLAB/version information
process_trace.json	Per-stage process trace for the ordinary run
diagnostics.md	Readable diagnostics summary
app_snapshot.png	Full UI state at export time
registered axes	PNG/PDF according to selected format; empty axes can be blank by design

Current exporter limitation. The generic export callback is called with app.Results and app.Config. SweepResult, ResearchResult, SurfaceStudy, SurfaceProfile and PlazaLadder are separate in-memory properties. Their registered figures are exportable because their axes are in the registry, and the UI snapshot preserves the visible table, but do not assume the generic run_state.mat/metrics.csv contains those study structs. For a publication-grade workflow, keep the study’s visible table/figures and, if needed, add a dedicated serialization path in code rather than claiming the bundle already contains it.

21. How the AI must reason from the results

Observed pattern	Required conclusion
Smoothed error $\approx$ Slices error; measured evaluations lower	Valid evidence of computational reduction at matched quality. Then inspect growth slopes to decide constant-factor vs scaling change.
Smoothed error lower; evaluations higher	Accuracy improvement bought with more work; not a cost reduction.
Smoothed evaluations lower; error/mode loss worse	Not a win at matched quality. Report tradeoff/failure.
Predicted sparse cost lower; measured evaluations unchanged	Representation is potentially cheaper, but current implementation has not realized the saving.
Surface low-rank but $\text{rank}(R_0) \approx \text{rank}(R_1)$	Compactness may be inherited from terminal/fusion factors; do not credit smoothing alone.
Control axis changes surface spectrum	Diagnostic inconsistency: stop interpretation of E2 until the measured matrix/path is verified.
Four Doors RMSE improves while modes collapse	RMSE is flattering the failure. Use mode-weight/marginal diagnostics as primary evidence.
Plaza landmark mean far from truth while samples form reflected modes	Do not equate mean RMSE with posterior failure; inspect the distribution and range-consistent modes.

Observed pattern	Required conclusion
Grid World one layout fails at a certain variable count	Report a layout/mission-specific turn, not a universal engine limit.
Slices = Smoothed on a general-engine case	Mechanism may not have been exercised; not evidence that smoothing is useless/equivalent.

21.1 Minimum claim language

- **Allowed:** “On the Two-pose Lemma-1 route, Smoothed Slices used X% fewer measured factor evaluations at comparable relative-L1 error for this scenario/seed/budget.”
- **Allowed:** “The fitted work-vs-budget slopes are similar; the observed difference is consistent with a constant-factor saving rather than a changed growth class.”
- **Allowed:** “The active-set cost model predicts a reduction, but the current implementation’s factor-evaluation counter did not realize it because the dense transition support was still evaluated before sparsification.”
- **Not allowed:** “Smoothed Slices is asymptotically faster” unless the measured growth evidence and theory support that statement.
- **Not allowed:** “Smoothed Slices beats Slices on Plaza” as evidence of RCS if the current general engine did not consult innerEstimator.
- **Not allowed:** “NF-iSAM has zero factor-evaluation cost” as a superiority claim; that is a units/semantics mismatch.

22. Stop / warning / failure handling

Situation	AI action
No methods checked + Run Selected	Check at least one method; do not substitute Run All unless all three are intended.
Stop pressed during single run	Keep completed method results; label the run stopped/cancelled. Do not call missing methods failures.
Stop pressed during sweep	Finished cells are retained. Interpret only completed cells and preserve cancellation status.
Surface study says no surface built	Wrong route/build. Do not force a surface interpretation on a general-engine case.
Engine warning present	Copy it verbatim into the record; inspect whether it affects support, retained mass, training, or posterior reliability.
Figure checklist says empty	Run the corresponding tab if that figure is required; otherwise accept blank export as intentionally not generated.
A later build changes stage numbering	Identify stage by eliminated variable/process card, not by historic numeric index.
Results look unexpectedly identical	Check whether the current engine actually consulted innerEstimator before concluding equivalence.
appFast is still selected	Switch to paperLike and rerun before reporting any scientific number.

23. Mandatory experiment record for the AI

For every run intended to support a conclusion, write a record with these fields. Do not rely on memory or only on screenshots.

Field	Required content
Run ID	Unique label, e.g. E16-M_seed3_N200
Date / current archive	proj_zip2 / 17-Aug-2026 or later verified code revision
Case / variant / layout / poses	Exact UI values
Seed	Exact Reproducibility value; also record automatic sweep seed if applicable
Budget preset	paperLike / appFast + any manual overrides
Visible cardinalities	X0 , Ln , S , X1 , N0 , backward samples, marginal grid

Field	Required content
Replay	OFF for reference; otherwise cache and early-stop states
Methods	Exactly which methods actually ran
Status/warnings	Per method; verbatim warning text
Posterior quality	Relative L1, MMD, RMSE, mode metrics as case allows
Measured work	Factor evaluations and other counters where semantically comparable
Surface diagnostics	Rank/spectrum/sparsity/K/retained mass if surface is actually built
Runtime	Secondary practical metric; include machine/release context
Interpretation	One sentence constrained by Section 21 decision rules
Export path	Folder containing figures/config/diagnostics

Appendix A. Automatic plans hidden behind the buttons

These values are encoded by the current functions and are not user-editable from the app's main fields during the corresponding study.

Button / plan	Automatic design
Research Question → Measure cost vs quality	Two-pose only; d=2.2; 4 scenarios; N={50,100,200,400}; seed=3; Slices + Smoothed Slices only.
Sweep → Two-pose range: budget and ambiguity	6 scenarios x 4 budgets = 24 cells; separations 1.2/2.2/3.5; tight odom sigma 0.6, weak 3.0; seed=3; all three methods.
Sweep → Four Doors: spacing and odometry	6 scenarios x 4 budgets = 24 cells; spacing 2.5/4/6, weak sigma 0.55, tight 0.18, irregular and long-corridor scenarios; seed=3; all three.
Sweep → Grid world: problem size	18 cells at N=200, backward=200, S =250; base seed 11, extra seeds 3/7/13; office/corridor/warehouse; sensing/association families; all three.
Surface Complexity → Measure surface complexity	17-run one-axis screening; X0 =120, X1 =120, S =101, backward=100, MMD eval=100, K=24; control/sampling/fusion/overlap axes.
Surface Complexity → Profile the active set	Two-pose multimodal; X0 =120, X1 =120, S =101; K={2,4,8,16,32,64,128,∞}; surface tolerance=1e-3.
Plaza → Run evidence ladder (P2-A)	Three independent windows chosen from actual landmark evidence; dataset = currently loaded Plaza1/Plaza2; landmark = spinner; methods = Compare Methods checkboxes; case-window seed defaults to 7 unless code is changed.

Appendix B. Current GUI boundaries — do not invent these controls

- **No GUI selector for RCS-Basis, RCS-Regression, RCS-Proposal, or RCS-Hybrid.** Current operational method is finite-support RCS Mode A.
- **No GUI field for NF-iSAM training samples or max iterations.** They move through budget preset: paperLike=2000/200, appFast=500/60.
- **No GUI field for Two-pose mixture separation.** The ordinary multimodal case uses d=2.2 automatically; sweeps vary it internally.
- **No GUI slider for Plaza ambiguity fraction.** variant=multimodal maps to the builder's default ambiguous-association fraction (0.5). The full P1-B fraction sweep is script-level.
- **No GUI buttons for Plaza P1-B or P2-B.** The only dedicated Plaza protocol button in this build is Run evidence ladder (P2-A).
- **No separate “small range-only SLAM” case label.** The current app exposes Grid world and Plaza cases instead.
- **No Implementation Notes tab.** It was intentionally removed; equations/provenance/checklists live in code/data/tests.

Appendix C. One-page operating checklist

- ☐ Launch runSmoothedSlicesApp and verify the 11 current tabs.
- ☐ Smoke: Seed 0 → Two-pose Gaussian → appFast → Run All → mark NOT FOR REPORTING.

- ☐ Scientific reset: Seed 3 → Two-pose Gaussian → paperLike → replay OFF → Run All.
- ☐ Repeat Two-pose with variant multimodal; keep every budget identical.
- ☐ Process Explorer: $k=1$, stage=2/11 → capture Slices and Smoothed frames.
- ☐ Research Question: restore paperLike unswept fields → Measure cost vs quality → inspect every scenario.
- ☐ Surface Complexity: Measure surface complexity → cycle all axes → verify control.
- ☐ Surface Complexity: Profile the active set → record surface-K, posterior-K, predicted vs measured cost.
- ☐ Four Doors: paperLike single run + Four Doors sweep → mode metrics before RMSE.
- ☐ Grid World: seed 11 office known baseline + ambiguous partner + problem-size sweep; report per-layout behavior.
- ☐ Plaza: seed 7, paperLike with $|X_0|=150$ → Plaza1 baseline → Plaza2 baseline.
- ☐ Plaza: $L=2$, all three checked → Run evidence ladder (P2-A) → state independent-window caveat.
- ☐ Use incremental replay only after the batch reference, as a separate labelled study.
- ☐ Diagnostics: copy every warning; do not hide failed/cancelled methods.
- ☐ Figure Generator: PNG and PDF → Refresh checklist → verify drawn/empty → Export bundle.
- ☐ Write the experiment record and constrain conclusion using Section 21.

Appendix D. Source-of-truth files used to construct this guide

File	Authority
src/SmoothedSlicesAppProgrammatic.m	Exact current tabs, controls, callbacks, button behavior, export wiring
src/+methods/commonMethodConfig.m	Default paperLike sampling/RCS/MMD/incremental settings
src/+methods/configPreset.m	paperLike/appFast grouped budgets and hidden NF-iSAM training settings
src/+methods/twoPoseSweepPlan.m	Two-pose sweep scenarios, budgets, seed and controlled axes
src/+methods/fourDoorsSweepPlan.m	Four Doors sweep design
src/+methods/gridWorldSweepPlan.m	Grid World structural sweep design
src/+research/surfaceComplexityStudy.m	E2 screening design and control logic
src/+research/activeSetProfile.m	Active-K profile, dense baseline and predicted-vs-measured cost logic
src/+datasets/makeTwoPoseRangeCase.m	Exact Eq.-16 benchmark, variants and elimination route
src/+datasets/makePlazaCase.m	Windowed Plaza modeling assumptions and default data settings
src/+experiments/runPlazaProtocol.m	P2-A ladder semantics and paper-style Plaza protocol defaults
README.md and current project reports	Research framing, implementation limitations, interpretation cautions

END OF OPERATOR GUIDE